

Lifetime Asset Allocation in New Zealand: A Simulation and Optimization Approach

Callum Davidson

May 17, 2026

1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (hereafter, Consilium). Consilium is a New Zealand-based financial services company that provides tools and consulting services to financial advisers and their clients in areas including investment management, portfolio construction, risk management, and financial planning. Consilium also offers managed funds and portfolio solutions designed to support advisers in meeting a wide range of client needs. The aim of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with particular focus on retirees in the New Zealand context. Our objective is to produce insights and recommendations that may help Consilium better support advisers and their clients.

Although this research is conducted in partnership with Consilium, the views expressed in this paper are ours and do not necessarily reflect the official policy or position of Consilium. We retain full academic independence in the design, analysis, and reporting of this work. We also retain full ownership of the intellectual property associated with this research, while granting Consilium a non-exclusive licence to use the findings internally and share them with clients and partners. No confidential or proprietary information provided by Consilium has been used in this study.

We thank Consilium for their support and collaboration in this research project, and for the feedback that has helped shape the direction and focus of this work.

Contents

1 Foreword and Research Context	1
2 Introduction	2
2.1 The Glidepath in Academia	3
2.2 Problem Statement and Research Questions	6
2.3 Our Approach and Specification	7
3 Retirement in New Zealand	7
3.1 New Zealand Superannuation Policy Settings	8
3.2 Patterns of Spending in Retirees - New Zealand and Abroad	9
3.2.1 Spending By Age	9
3.2.2 Bequests and Precautionary Savings	10
3.2.3 Spending by Household Composition and Wealth	11
3.2.4 Expenditure Benchmarks for New Zealand Retirees	11
4 Methods for Simulating and Optimising Lifetime Wealth Trajectories	12
4.1 Lifecycle Simulation Framework	13
4.1.1 Modelling Contributions and Withdrawals	14
4.1.2 Modelling Asset Allocation Policies	16
4.2 Methods for Simulating and Bootstrapping Asset Returns	18
4.2.1 Block Bootstrap Methodology	18
4.2.2 Data for Block Bootstrapping	19

5	Optimal Lifetime Asset Allocation Strategies for New Zealand Retirees	20
5.1	Optimal Age-Based Glide-Paths	20
5.2	Optimal Wealth-Responsive Allocation Strategies	20
6	Discussion and Implications	20
7	Conclusion	20
A	Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)	23

2 Introduction

Lifetime asset allocation is one of the highest-impact financial decisions households make.

In general, sources of higher asset returns are associated with higher risks (e.g., (Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007)). This makes the task of selecting an optimal allocation strategy a complex balancing act, weighing the investor’s need for growth to fund future spending against their tolerance for volatility and potential losses. The problem is further complicated by the fact that very few of these factors are known in advance: future returns, future spending needs, and even the investor’s own risk tolerance can all evolve over time in response to changing circumstances.

Put simply, the problem of lifetime asset allocation is one of making the best possible decisions in the face of deep uncertainty about the future. The answer to this problem matters at both household and fiscal levels. The global asset-backed pension market is large (OECD, 2025), and in New Zealand retirement outcomes directly interact with public finances through New Zealand Superannuation. Treasury projections indicate substantial long-run pressure from population ageing (The Treasury, 2025), while policy and industry commentary increasingly emphasises higher private saving and better KiwiSaver decisions (Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025). Better allocation policy therefore has private welfare value and public-finance relevance.

Although many investors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, allocation to these broad asset classes is the primary driver of a portfolio’s long-term behaviour. This is even more true for typical retirement portfolios (and especially products like KiwiSaver) which are usually held in managed funds. In New Zealand, KiwiSaver Schemes typically do not allow the investor to exercise granular control over security selection, but rather give them a choice between a small number of pre-defined strategies allocating to different amounts of “Growth” and “Income” asset classes (Financial Markets Authority, 2015). So called “Growth” assets typically comprise a mix of domestic and international equities, while “Income” assets are a mix of domestic and international bonds and cash. In this context, the key decision for most investors is not which specific stocks or bonds to hold, but rather how much of their portfolio to allocate to Growth versus Income assets, and therefore how much risk to take on.

A well performing allocation strategy is one that maximises the investor’s expected welfare, which is a function both of the expected return and the protection against adverse outcomes. Where the investor’s needs are fixed, Merton (1969) shows that the optimal risky-asset share is constant through time. There are many simplifying assumptions in this result, but probably the most significant for lifecycle investing is that the investor’s spending needs are fixed and known in advance. In reality, this is not true as differential longevity risk, health shocks, and changing preferences all mean that spending needs are uncertain and evolve over time.

Broadly speaking, the investor’s lifecycle can be divided into two phases: accumulation and decumulation. During the accumulation phase, the investor is typically working and saving, building up their wealth over time. When they transition to retirement, they enter the decumulation phase, where they draw down on their accumulated wealth to fund their spending needs. This report will focus primarily on the decumulation phase, but the accumulation phase is also relevant for understanding the optimal trajectory of risk over the lifecycle, and so we will spend some time discussing the literature on both phases, especially as it relates to the shape of the optimal glidepath (the trajectory of risk over time).

While Merton (1969) deals specifically with fixed spending needs in decumulation, **modigliani1954lifecycle** develops a more general lifecycle model that incorporates both phases and the transition between them. The Life Cycle Hypothesis (LCH) posits that rational households smooth consumption over a finite lifetime by saving during their earning years and drawing down wealth in retirement, with the trajectory of wealth following a characteristic hump shape that peaks around the point of retirement. This model, expanded several times

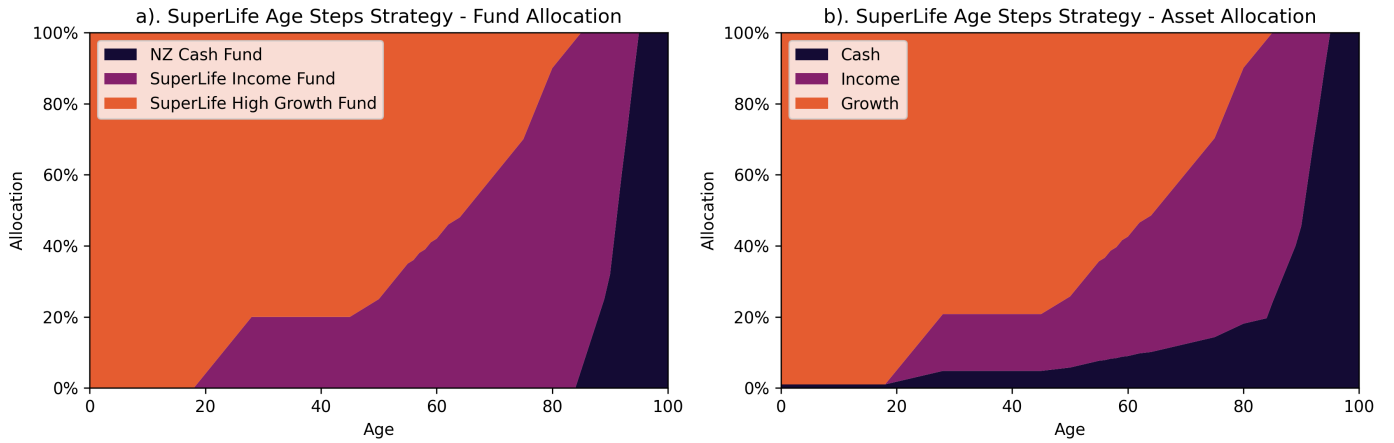


Figure 1: SuperLife’s Age Steps target strategy as per their Statement of Investment Policy and Objectives (SIPO), showing the allocation to different funds (a) and the resulting allocation to different asset classes (b). Source: SuperLife (2026).

by Modigliani and others (see Modigliani, 1986) provides a enduring theoretical framework for understanding the lifecycle of saving and spending, which is a topic this report will return to in the context of understanding the optimal trajectory of spending and risk over the lifecycle.

With respect to asset allocation across the lifecycle, Bodie et al. (1992) adds in the concept of human capital, which represents the present value of future labor income. When human capital is bond-like, younger households can rationally hold more equity in financial wealth and reduce equity with age (insofar as they continue to be employed). This gives the main theoretical foundation for the intuitive idea that younger investors should hold a higher proportion of their portfolio in risky, high-return assets, and then gradually reduce this proportion as they approach retirement age. This informs the intuitive principles for conventional wisdom around investing over the lifecycle. That is, in essence, that younger investors should take more risk because they have more time to recover from potential losses, while older investors should take less risk to preserve their accumulated wealth as they approach retirement and begin drawing down on their assets. Based in a understanding of these principles, a large number of intuitively appealing, easy-to-communicate age-based rules have emerged that have become the industry standard for retirement investing. These include the various age-in-bonds rules (e.g. the 120-minus-age rule), and target-date funds which are investment products that automatically adjust the asset allocation over time according to a predetermined schedule. Collectively, these types of strategies where asset allocation is a deterministic function of age are often (and hereafter) referred to as “glidepaths”. A great deal of work has been committed to studying the optimal shape of these glidepaths, both in terms of their theoretical properties and their empirical performance. They have been developed into a myriad of financial products and are widely recommended by financial advisers and retirement planning tools. A local example is the SuperLife Age Steps strategy, which automatically adjusts the allocation to Growth and Income assets based on the investor’s age, with the proportion of Growth assets declining as the investor gets older (see Figure ??).

Although widely recommended, the optimal shape of the glidepath is a subject of active debate in the literature, with different studies reaching different conclusions about whether risk should decline, rise, or remain constant over time, and whether the optimal strategy differs between the accumulation and decumulation phases. The following section will review the literature on this topic, with particular attention to where disagreements arise and how they are driven by modelling choices rather than fundamental empirical disagreement. We will then outline our research questions and approach for addressing these questions in the New Zealand context.

2.1 The Glidepath in Academia

What emerges from the literature is a complex and unresolved debate about the optimal shape of the glidepath, with different studies reaching different conclusions about whether risk should decline, rise, or remain constant over time, and whether the optimal strategy differs between the accumulation and decumulation phases.

There are essentially five main shapes that have been proposed for the optimal glidepath (see Figure ??):

- Declining equity (DE): the conventional wisdom that risk should decline with age, often implemented as a simple age-based rule.

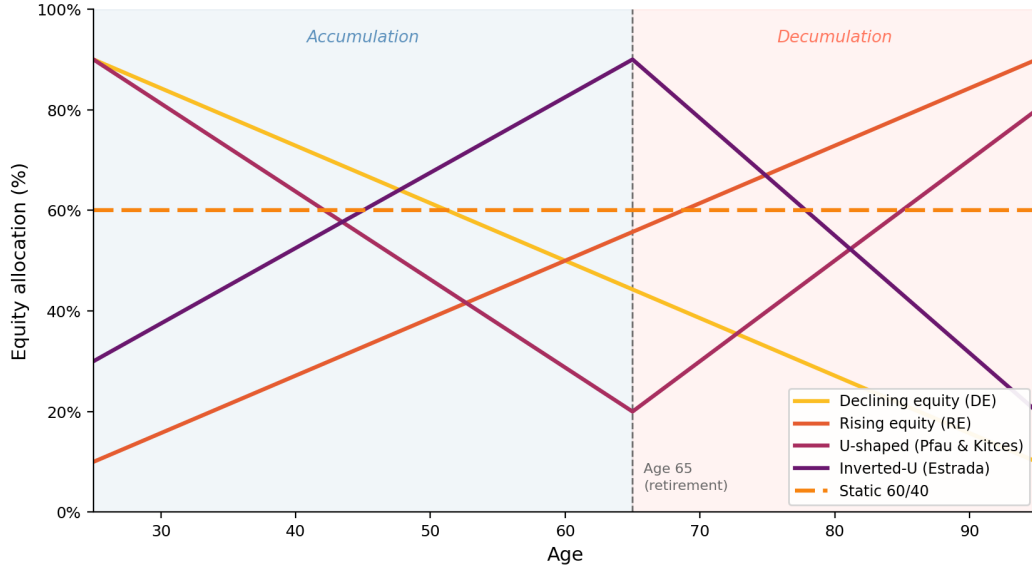


Figure 2: Illustration of different proposed glidepath shapes. The x-axis represents age, with the vertical dashed line indicating the point of retirement. The y-axis represents the equity share of the portfolio. The different lines represent the different proposed glidepath shapes: declining equity (DE), rising equity (RE), U-shaped, inverted U-shaped, and static allocation.

- Rising equity (RE): the contrarian view that risk should actually increase with age, particularly at the start of retirement, to capture the higher expected returns of equities when they are most needed.
- U-shaped: declining equity during the accumulation phase, followed by rising equity during retirement.
- Inverted U-shaped: rising equity during the accumulation phase, peaking at retirement, followed by declining equity during retirement.
- Static allocation: the view that a constant equity share throughout life is optimal, and that the shape of the glidepath is less important than the overall level of risk taken.

The reasons for this disagreement are complex, but a key fault line in the literature is the choice of methodology for generating return scenarios and evaluating strategies. Different studies use different approaches, including historical rolling periods, Monte Carlo simulation with IID returns, and block bootstrap methods that preserve time-series dynamics. These methodological choices have a significant impact on the results and conclusions drawn about the optimal glidepath shape.

The standard industry recommendation is that investors decrease their risk exposure as they age, continuing to reduce risk even after retirement. This approach appears in investment textbooks (e.g Bengen, 1996; Bodie et al., 2023) and is widely recommended by financial advisers and retirement planning tools (Choi, 2022). A review of U.S target date funds (TDFs) by Dolvin et al. (2010) find that most follow some form of declining-risk structure, often resembling a 120-minus-age equity rule. For a domestic example, take the SuperLife Age Steps allocation profile shown in Figure ???. At one point a declining-risk glidepath was considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (see R. Shiller, 2005; R. J. Shiller, 2006).

However, declining-risk glidepaths have been challenged by a substantial body of academic work, and the post-retirement period in particular is a site of active and unresolved disagreement in the literature. Responding to the aforementioned Bush-era proposal, R. Shiller (2005) and R. J. Shiller (2006) find that declining-risk glidepaths often fail to outperform a 100% equity strategy. Estrada (2014) and Estrada (2015) find that declining-risk strategies tend to underperform constant-risk alternatives or even contrarian rising-risk strategies across a wide range of countries and historical periods.

Studies generating return scenarios using Monte Carlo simulation with IID returns tend to find support for U-shaped glidepaths. This means a declining-risk strategy during the accumulation phase, equity exposure low around the point of retirement, and then rising equity during the retirement phase. The rationale is that starting conservatively limits the damage from a bad early period (which is particularly harmful at the point of retirement when the portfolio is largest), while systematically accumulating equities at depressed prices through a downturn allows the portfolio to capture the eventual recovery when growth is most needed. The

proposed risk mechanism that this avoids is a sequence-of-returns argument, that is the risk of a series of bad returns early in retirement that depletes the portfolio when it is largest and most vulnerable. In this line of thinking, the accumulation phase is more supportive of a DE strategy, because the sequence risk mechanism is less relevant when the portfolio is still growing and not yet being drawn down. Here the investor can afford to be more aggressive early on, capturing the higher expected returns of equities when they have the longest time to compound, and then gradually reduce risk as they approach retirement and peak wealth. This is the conventional lifecycle logic that underpins most age-based glidepaths. Proponents of a U-shaped glidepath argue that the same logic applies in reverse during retirement: starting conservatively at the point of retirement limits the damage from a bad early period, while systematically accumulating equities at depressed prices through a downturn allows the portfolio to capture the eventual recovery when growth is most needed.

Pfau and Kitces (2014) make the case for rising equity during retirement using Monte Carlo (MC) simulation. Across 10,000 simulated retirements with lognormal IID returns, 121 glide path combinations (varying start and end equity from 0% to 100%), and both 4% and 5% withdrawal rates, they find that rising equity during retirement reduces both the probability and magnitude of portfolio failure. The optimal starting allocation is 20–40% equity, rising to 60–80% by end of retirement. A 30%→60% rising path beats a static 60/40 on failure rate *and* failure magnitude. This holds across three capital market assumption sets. They argue that retirement outcomes are almost entirely determined by the first 15 years. A RE strategy during retirement starts conservative, limiting the damage from a bad early period (and at peak asset accumulation), and then systematically accumulates equities at depressed prices through the downturn.

Anarkulova et al. (2025) use a block-bootstrap methodology applied to a dataset of monthly real returns across 39 developed countries from 1890 to 2023, drawing on over 31,800 country-months of data and preserving the time-series and cross-sectional structure of historical returns. Under block bootstrap, the optimal allocation is approximately 100% static equity throughout the lifecycle. The sole exception is a brief tactical allocation to bills at the point of retirement, which does resemble a kind of aggressive U-shaped glidepath, but they argue that this is a special case, driven by the rigidity of a fixed real 4% withdrawal rule (based on Bengen, 1994) and when spending flexibility is restored, this exception disappears and the all-equity result holds at every age. They also argue that the conventional preference for bonds is substantially overstated by return-generation decisions in the literature. They compare optimal glidepaths under IID return sampling and block bootstrap, and find that the optimal bond allocation is higher under IID assumptions. They argue that this is because the long-horizon properties of bonds are much less favourable than their short-horizon statistics suggest. At 30-year horizons, bond variance *increases* relative to the one-year level (variance ratio 2.30), whereas equity variance *decreases* (variance ratio 0.75). Bonds also carry a correlation of -0.78 against inflation, which they argue destroys real purchasing power in precisely the environments most harmful to retirees.

In essence, Anarkulova et al. (2025) argue that IID assumptions overstate the risks of sequence-of-return risk. In reality, timeseries effects mean the sequence is actually autocorrelated and mean-reverting. In this environment, the higher expected returns of equities are more reliably captured over long horizons and the risks of bonds are more pronounced.

Using deep reinforcement learning to solve a comprehensive lifecycle model against Monte Carlo simulated returns (with autocorrelated return dynamics), Duarte et al. (2021) find the optimal equity allocation at retirement is approximately 60%, and that TDFs (which land at 30–40% equity) are too conservative; the welfare cost of following a TDF rather than the optimal policy is 1.7–2.8% of lifetime consumption. However, they otherwise find that a declining trajectory of equity allocation proposed by many TDFs (declining equity during accumulation) is broadly optimal, and that the optimal trajectory post-retirement is approximately flat (not declining or rising).

However, there is considerable scepticism toward conventional declining-equity strategies during both the accumulation and decumulation phases. The argument is essentially that a DE strategy lowers the expected return of the portfolio precisely when the portfolio is at its largest: lifecycle funds are aggressive when the portfolio is small (early career) and conservative when it is large (near retirement), which is backwards. Estrada (2014) establishes this for the accumulation period. The companion paper, Estrada (2015), applies the same DMS dataset to the retirement drawdown period, testing 81 rolling 30-year retirement windows starting from 1900 with a 4% real withdrawal rule. The key finding reverses the accumulation result: DE strategies have lower failure rates *and* higher terminal wealth than rising-equity mirrors in every specification. In the United States, DE failure rates range from 4.9 to 8.6%, while the corresponding RE mirrors fail at rates of 8.6 to 21.0%. Across all 19 countries, DE fails at 31–32% versus RE at 39–49%. The mechanism is a sequence-of-returns argument specific to the drawdown context. A retiree’s portfolio is largest at the point of retirement; DE starts equity-heavy at this point, capturing equity gains while the portfolio is large and compounding on a large base. As the portfolio shrinks through withdrawals, the gradual shift to bonds provides capital preservation. RE

strategies invert this: equity exposure builds only after withdrawals have already depleted the portfolio, so the eventual equity gains arrive too late and on too small a base to matter. Combined with the 2014 accumulation findings, Estrada’s recommended full-lifecycle shape is an *inverted U*: equity should increase during working life, peak at the point of retirement, and then decline through the drawdown period.

The DE position is further weakened by evidence that even dynamic DE management adds little or no value over a well-chosen static allocation. Blanchett (2007) tests 43 glide paths (11 static allocations and 4 types of declining-equity dynamics (linear, stair-step, concave, and convex)) bootstrapped from 80 years of monthly return data, across 1,071 scenarios defined by distribution period (20–40 years) and real withdrawal rate (3–8%). Rising-equity paths are not tested; the comparison is strictly static versus DE. Static allocations have the lowest failure probability in all but one of the 1,071 scenarios. The best static portfolio by raw failure probability is 100% equity; using his “Success to Variability Ratio” (probability of success divided by portfolio standard deviation), the best risk-adjusted portfolio is a 60/40 stocks-bonds allocation.

An important contextual observation from Blanchett (2007) is that the choice of glidepath matters little when withdrawal rates are moderate and horizons are short. At a 4% withdrawal rate over a 20-year period, the range of failure probabilities across all 43 strategies is only 1.56 percentage points. The same metric reaches 62.45 percentage points at a 6% withdrawal rate over 40 years. The glidepath debate is therefore most consequential for aggressive scenarios, and the literature’s disagreements should be interpreted in that light.

The tension between these results is not easily resolved, but it is clear that the choice of return-generation methodology is a key driver. Additionally, the choice of risk measure (failure probability, expected shortfall, risk-adjusted success ratio, expected utility) and the specification of the spending rule (fixed real withdrawal, flexible spending, etc.) also matter. Pfau and Kitces (2014) use Monte Carlo with independently and identically distributed (IID) lognormal returns, in which bad returns are equally likely at any point and carry no autocorrelation.

Historical rolling periods (Estrada, Bengen) preserve the actual structure of return sequences. They argue that because recessions cluster, valuations mean-revert, and bad periods tend to be followed by recoveries, IID assumptions overstate the risks of sequence-of-returns risk as stocks are more likely to recover from an early dip than the IID model suggests. This approach does, however, limit the sample size to the number of non-overlapping historical periods, which is a particular problem for long-horizon retirement planning where the number of 30- or 40-year periods in the historical record is very small. Anarkulova et al. (2025) use a block bootstrap approach, in which 6–12 month blocks of returns are sampled from the historical record and concatenated to create synthetic return paths. This preserves some of the time-series dynamics of returns, including mean reversion and volatility clustering, while generating a much larger sample of data. However, even in this approach it is unclear how “novel” return paths are, particularly given the small number of downturns even in a long historical record. This means that the optimisation could be overfitting to a small number of historical downturns with the particular time-series dynamics that they exhibit. Duarte et al. (2021) use a Monte Carlo approach with autocorrelated return dynamics, which might arguably be more realistic than IID assumptions, but still relies on a parametric model of return dynamics that may not capture all relevant features of the data. Interestingly, they find some support for the typical DE trajectory during the accumulation phase (typically recommended by TDFs), but not during the retirement phase, where they find a flat trajectory is optimal.

The choice of objective function also matters. Pfau and Kitces specifically target failure risk minimisation; Estrada targets both failure risk and terminal wealth jointly and finds DE dominates on both. Blanchett introduces the risk-adjusted Success to Variability ratio and finds it tilts the optimal allocation toward moderate equity, regardless of glidepath shape. Studies optimising expected utility with explicit risk aversion parameters (e.g. Duarte et al. (2021)) tend to find approximately constant, high equity allocations consistent with neither the U-shape nor the inverted-U, but broadly aligned with Blanchett’s static recommendation.

Finally, the specification of the investor’s spending and income needs is a critical driver of optimal policy. In discussing the paper by Anarkulova et al. (2025), Felix (2025) argues that the real risk to investors is not the risk of portfolio failure per se, but the risk of being unable to adjust spending in response to market conditions. For our purposes, investor spending should be grounded in realistic benchmarks of actual spending patterns in retirement, which are not necessarily well captured by a fixed real withdrawal rule.

2.2 Problem Statement and Research Questions

The review above reveals that the literature has not converged on a single answer to even the most basic question of optimal retirement allocation, and that disagreements are substantially driven by modelling choices rather than fundamental differences in the underlying economic problem. The choice of return-generation methodol-

ogy (IID Monte Carlo, historical rolling periods, or block bootstrap), the choice of risk or objective measure (failure probability, terminal wealth, expected utility), and the specification of the spending rule (fixed real withdrawal vs. flexible spending) each independently shift the apparent optimal policy. A secondary but practically important question is whether the glidepath shape matters at all compared with the more fundamental question of whether bonds should be held in retirement portfolios at the levels currently recommended.

This thesis addresses the following core problems in the New Zealand context: *(i) What lifetime allocation policy is most appropriate for New Zealand retirees under realistic return dynamics and spending paths? (ii) How sensitive are these recommendations to the modelling choices that drive disagreement in the literature? (iii) Can allowing allocation policy to respond to both age and wealth improve outcomes beyond what age-only policies achieve?*

Concretely, we ask:

1. Are conventional age-based declining-risk glidepaths optimal under standard assumptions, and does that result survive under NZ-calibrated return dynamics and spending paths that decline in real terms with age?
2. How sensitive are “optimal” recommendations to modelling assumptions—methodological (return-generation process, IID vs. block bootstrap, risk measure, spending rule) and substantive (household composition, housing tenure, real spending trajectory, longevity horizon, NZ Super as a guaranteed income floor)?
3. Does allowing allocation policy to depend on both age and wealth improve outcomes relative to age-only glidepaths, and if so, under what conditions?

2.3 Our Approach and Specification

We make three contributions. First, we synthesise the lifecycle allocation literature with explicit attention to where theory, empirical evidence, and industry implementation diverge—and where that divergence is attributable to modelling choices rather than genuine empirical disagreement. Second, we calibrate retirement spending to New Zealand institutions: we use empirically grounded spending trajectories that decline in real terms after retirement rather than a fixed real withdrawal rule, and we treat NZ Super as a guaranteed income floor that materially reduces the required drawdown from financial assets. Third, building on Mäkinen and Toivanen (2024), we implement a flexible Monte Carlo optimisation framework that can evaluate both age-based glidepaths and wealth-responsive policies across alternative return processes and risk measures.

The framework is intentionally modular: allocation policy class, spending rule, risk objective, and return model can each be varied independently. This is necessary because, as the preceding review makes clear, the disagreement in the literature is not about a single parameter but about the structure of the model itself. Our goal is therefore not to produce one universal glidepath recommendation; it is to provide a defensible, testable method for identifying which modelling choices drive conclusions, and to characterise the policy-relevant strategies that are robust across plausible specifications in the New Zealand setting.

3 Retirement in New Zealand

The New Zealand-specific literature on retirement concerns remains limited, so we draw extensively on international evidence. Understanding the institutional context is essential for interpreting this evidence appropriately. New Zealand’s retirement system combines a universal public pension (New Zealand Superannuation) with voluntary private savings through KiwiSaver. New Zealand Superannuation provides all residents aged 65 and over with a baseline income regardless of work history or assets, and adjusts annually for inflation. KiwiSaver allows voluntary contributions from employees, matched by employers and the government. Together, these mechanisms provide a safety net while encouraging personal saving (Ministry of Social Development, 2022; Te Ara Ahunga Ora: Retirement Commission, 2021).

KiwiSaver was introduced in 2007, so most current retirees have little or no accumulated balance. Currently, 40 percent of New Zealanders aged 65 rely entirely on New Zealand Superannuation, with a further 20 percent holding minimal additional savings (Retirement Commission Te Ara Ahunga Ora, 2022). As the scheme matures and government policy encourages higher contribution rates (Luxon, 2025), future cohorts will retire with substantially larger balances. This demographic shift increases the need for robust guidance on asset allocation and withdrawal strategies during retirement.

Given these trends, the Australian experience becomes increasingly relevant. Australia’s retirement system combines a means-tested public pension with mandatory superannuation contributions of 12 percent of income

(Australian Taxation Office, n.d.; Swoboda, 2014). While institutional differences remain, the growing role of KiwiSaver makes the two systems more comparable than in the past.

3.1 New Zealand Superannuation Policy Settings

New Zealand Superannuation (NZ Super) is a universal, non-means-tested public pension paid from age 65, funded from general taxation rather than a contributory insurance scheme. It forms the guaranteed income floor underpinning all retirement planning in New Zealand. Entitlement is not conditional on prior employment, investment, or KiwiSaver participation; a recipient’s private wealth or income has no effect on the payment rate.

Eligibility

The primary eligibility criteria are age and residency. A person becomes entitled upon turning 65, provided they have been both resident and present in New Zealand for a sufficient period since age 20. For those born on or before 30 June 1959, the requirement is 10 years total, of which 5 must have accrued since age 50. For those born on or after 1 July 1977, the total requirement is 20 years (with 5 since age 50), reflecting a graduated increase introduced by the Social Assistance (Residency Qualification) Legislation Act 2018. Special provisions apply to recognised refugees and to periods spent in the Cook Islands, Niue, and Tokelau (New Zealand Parliament, 2001, s. 8).

Payment Rates

Rates are set in Schedule 1 of the New Zealand Superannuation and Retirement Income Act 2001 and adjusted each 1 April by Order in Council. The April 2026 rates (before deduction of tax under code M) are listed in Table 1.

Table 1: NZ Super rates from 1 April 2026 (before tax). Source: Work and Income New Zealand (2026a, 2026b).

Rate type	Per week	Per fortnight	Per year
Single living alone	\$647.37	\$1,294.74	\$33,663
Single sharing	\$595.57	\$1,191.14	\$30,970
Couple (per person)	\$492.14	\$984.28	\$25,591
Couple (combined)	\$984.28	\$1,968.56	\$51,182

After deduction of standard tax (code M), the fortnightly after-tax rates are approximately \$1,110 for a single person living alone and \$1,708 combined for a couple, equivalent to roughly \$28,900 and \$44,400 per year respectively (Work and Income New Zealand, 2026a, 2026b). These figures differ slightly from those used in Matthews (2025) (who report \$519 and \$799 per week for singles and couples respectively) due to differing tax rate assumptions; this study uses the WINZ statutory rates as the primary reference.

Indexation and the Wage Floor

The Act mandates annual adjustment for CPI inflation (so the rate cannot decline in real terms) subject to a binding wage floor. The combined after-tax couple rate must remain between 66 and 72.5 percent of net average ordinary-time weekly earnings, as measured by the Quarterly Employment Survey (New Zealand Parliament, 2001, s. 16). In practice, wages have grown faster than CPI in recent decades, meaning the wage floor has been the binding constraint. The single living alone rate is fixed by statute at 65 percent of the couple combined rate (after tax), and the single sharing rate at 60 percent. This legislated floor is a key feature distinguishing NZ Super from defined-benefit pension schemes that index only to prices.

Policy Adjustments and Reform Debate

Despite ongoing fiscal pressure from population ageing, NZ Super’s core parameters have remained largely unchanged since 2001. The most significant recent amendment was the gradual increase in the residency qualifying period from 10 to 20 years for those born after 1 July 1977, phased in from 2018. The eligibility age of 65 has not changed; a proposal to raise it to 67 was advanced and subsequently abandoned. A 2022

government review (the Retirement Review Issues Paper, conducted by Te Ara Ahunga Ora) examined a range of possible adjustments including age, means-testing, and residency, without proposing binding changes (Ministry of Social Development, 2022).

Supplementary Benefits

Several additional transfers are available to eligible retirees:

- **Winter Energy Payment:** a universal supplement paid automatically to NZ Super recipients from 1 May to 1 October each year. The 2026 rate is \$20.46 per week for single people and \$31.82 per week for couples and people with dependants (Work and Income New Zealand, 2026a). No application or means test is required.
- **Accommodation Supplement:** a means- and asset-tested housing cost subsidy available to renters, boarders, and mortgage holders across all age groups including retirees. The amount varies by accommodation costs and location zone, with higher rates in Auckland and Wellington. Eligible retirees with moderate housing costs may receive up to several hundred dollars per fortnight (Work and Income New Zealand, 2026a).
- **SuperGold Card:** a concession card issued automatically to NZ Super recipients, providing discounts on public transport (including free off-peak travel on most regional services), selected retail, and other services (Ministry of Social Development, 2022).
- **Residential Care Subsidy:** an asset-tested subsidy for long-term residential aged care (rest homes and dementia units). As at 2022, a single person is generally required to contribute their own assets up to a threshold of around \$240,000 before the subsidy takes effect, with the family home exempt for a defined period if a spouse, dependent, or carer remains in residence (Ministry of Social Development, 2022).

Together, NZ Super plus the Winter Energy Payment provide a reliable inflation-linked income floor for all retirees.

3.2 Patterns of Spending in Retirees - New Zealand and Abroad

3.2.1 Spending By Age

The Stats NZ Household Economic Survey (HES) provides the most comprehensive data on New Zealand retirement spending patterns. Two recent reports are particularly relevant. Le and Richardson (2023) analyse expenditure patterns by age across New Zealand households, while Retirement Income Interest Group (2024) focus specifically on spending through retirement and its implications for drawdown strategies.

Using HES data for the year ended June 2019, Le and Richardson (2023) find that expenditure declines sharply at retirement and continues to fall thereafter. Mean household expenditure is \$64,700 for ages 65–74, then 28% lower for ages 75–84, and 44% lower for ages 85+. Expenditure declines across most categories, with the largest falls in mortgage repayments, air travel, and recreation. Unusually, healthcare expenditure also declines with age in New Zealand, likely reflecting the public funding of most core medical services. Retirement Income Interest Group (2024) estimate the real decline at roughly 2% per year after age 65.

The HES data cannot, however, distinguish voluntary from involuntary spending reductions. Retirees may reduce expenditure because they become less active and have genuinely lower needs, or because they face binding financial constraints. Without longitudinal data tracking individuals over time, these explanations cannot be separated cleanly. The observed decline likely reflects both channels.

Understanding the source of this decline is central to evaluating welfare. If spending falls primarily due to financial constraints, optimal strategies should aim to sustain pre-retirement consumption levels. If it reflects voluntary lifestyle adjustment, forcing consumption to remain high may be neither necessary nor welfare-improving. This distinction also affects risk tolerance: financially constrained retirees may need safer portfolios, while those adjusting voluntarily may tolerate more risk.

As we have mentioned earlier, beginning with work by Modigliani and Brumberg (2005), and expanded several times by Ando and Modigliani (1963), and others, the Life-Cycle Hypothesis (LCH) establishes a theoretical or “ideal” pattern of saving and consumption over the life course. Under the LCH, individuals are expected to smooth consumption over time, in other words, they wish to essentially maintain a stable standard of living throughout their life. The LCH predicts that individuals will save during their working years,

accumulating wealth to fund consumption in retirement (but only saving insofar as it does not unnecessarily reduce their pre-retirement consumption) and then draw down that wealth during retirement to maintain their living standards, ultimately exhausting their wealth near the end of life. The widely observed decline in spending at retirement (and lower-than-expected wealth drawdown), therefore, is puzzling under the LCH framework, and has been termed the “retirement consumption puzzle” by various authors. Understanding the source of this decline has been the topic of numerous papers and many explanations have been proposed.

Proponents of the LCH might argue that the observed decline in spending at retirement is primarily driven by financial constraints, which compel retirees to reduce their consumption below their desired level. This could be due to insufficient savings, unexpected health shocks, or other factors that limit retirees’ ability to maintain their pre-retirement standard of living. However, the widely observed tendency of retirees to not only reduce spending but also to not draw down on their wealth suggests that the decline in spending may not be solely due to financial constraints. King (1982) in a survey of Canadian retirees finds that the majority of respondents did not only fail to draw down on their wealth, but many actually accumulated more wealth during retirement. This finding has been corroborated by a number of subsequent studies in many different countries and time-periods (e.g. De Nardi et al., 2010; Love et al., 2009; Palumbo, 1999). A paper by the Grattan Institute in Australia (Grattan Institute, 2019) finds that fewer than half of Australian retirees draw down wealth at all, and over 40% accumulate wealth during retirement. Moreover, it finds that most Australian retirees are not financially constrained: retirees report feeling more financially secure than working-age households.

So whether how much of the observed decline in spending at retirement is driven by financial constraints versus voluntary lifestyle adjustment is an open question, but it does not appear that financial constraints are the primary driver. Consumption expenditure is an imperfect proxy for welfare, particularly when higher spending is driven by medical needs or lower spending reflects voluntary lifestyle adjustment. While Barrett and Brzozowski (2010) find that households retiring unexpectedly (and are thus unprepared) due to adverse health shocks do experience sharper spending declines than those retiring as planned (which is consistent with involuntary constraints driving some of the observed decline), the majority of the observed decline may be voluntary. Retirees spend less on commuting, work clothing, and purchased meals (Aguilar & Hurst, 2008; Battistin et al., 2009), and may substitute market purchases with home production, reducing measured expenditure while maintaining consumption (Becker, 1965; Hurd & Rohwedder, 2003). These mechanisms can mask substantial cross-household heterogeneity in retirement outcomes. Consistent with this interpretation is that the largest expenditure declines observed in the HES data by Le and Richardson (2023) are in discretionary categories such as transport and recreation, while spending on essentials such as food and housing remains comparatively more stable.

Cross-country comparisons highlight the role of institutional context. Banks et al. (2016) find that U.K. retirees reduce real consumption by 2.2% per year versus 1.4% per year in the U.S. This divergence is driven largely by healthcare financing: U.S. retirees face high out-of-pocket costs that rise with age, while U.K. retirees benefit from comprehensive public coverage. When healthcare spending is excluded, consumption trajectories are more similar across countries.

Australian evidence also shows spending declines of 0.5–1% per year in real terms (Grattan Institute, 2019). Unlike New Zealand’s cross-sectional HES data, the Grattan Institute analysis uses longitudinal data tracking individuals over time. The decline persists in both cross-sectional and longitudinal samples, suggesting it reflects genuine behavioural change rather than cohort effects. As Retirement Income Interest Group (2024) note, the weight of evidence indicates that spending reductions primarily reflect voluntary adjustment to changing lifestyle needs rather than binding constraints. From our perspective, this means that it is reasonable to model retirement spending as declining in real terms with age.

3.2.2 Bequests and Precautionary Savings

An aspect of the “retirement consumption puzzle” that is particularly troubling for the LCH is that a large number of retirees die with substantial wealth intact. This situation is often described by financial planners as a kind of risk, that is that a retiree may outlive their wealth and have thus died “underconsuming” relative to their potential. A common explanation for this phenomenon is that retirees intentionally preserve wealth to leave bequests to their heirs (e.g. Hurd, 1987). In this view, retirees are not underconsuming but are instead optimally consuming while intentionally the “utility” of leaving a bequest to their heirs. However, the empirical evidence for intentional bequest motives is weak. Alonso-García et al. (2022) find that Australian and Dutch retirees have little to no intentional bequest motive outside of provision for a surviving spouse, and that the observed pattern of non-decumulation is more consistent with saving for uncertain future costs, such as aged care or health shocks. This motive is often referred to as a “precautionary” motive, and is distinct from a

bequest motive in that the utility of leaving a bequest is not derived from the act of leaving a bequest itself, but rather from the desire to avoid the risk of being unable to cover future costs.

Precautionary motives are well-supported empirically. De Nardi et al. (2010) show that exposure to uncertain health and long-term care costs explains much of the observed non-decumulation in the United States. Similarly, Banks et al. (2016) find that healthcare costs drive higher late-life spending among U.S. retirees relative to those in the U.K. with broader public coverage. New Zealand’s system is mixed: most core medical services are publicly funded, but long-term care such as assisted living and memory care can involve substantial out-of-pocket costs (New Zealand Government, n.d.). Retirement Income Interest Group (2024) conclude that precautionary motives in New Zealand are likely weaker than in the United States but stronger than in countries such as France and Germany with more comprehensive public provision, placing New Zealand alongside the United Kingdom in a mixed-funding model. However, aged care is a significant cost to

3.2.3 Spending by Household Composition and Wealth

Household composition and housing tenure are among the strongest predictors of retirement spending levels and subjective wellbeing in New Zealand. Le and Richardson (2023) find that single-person retired households spend around \$30,700 per year on average, compared with \$65,100 for couples. Notably, spending increases more than proportionately with household size: the average couple spends more than double the amount of a single-person household, implying weaker economies of scale (in New Zealand at least) than other studies, such as (Anarkulova et al., 2025), have suggested in other contexts. Le and Richardson (2023) attribute the disproportionate gap primarily to differences in discretionary spending: couples devote substantially more to transport and recreation and culture, the two largest discretionary categories. Single-person households living alone are also more likely to be renters or to carry a mortgage, further constraining their discretionary budget, while couples are more likely to own outright.

Homeownership status has a particularly pronounced effect on both spending and wellbeing. Among retired New Zealand households, approximately 66 percent own their home outright, 14 percent carry a mortgage, and 14 percent rent (Le & Richardson, 2023). Outright owners report the highest subjective wellbeing of any tenure group and spend around \$55,800 per year, of which housing costs form a relatively small share. Renters spend considerably less in total, at around \$41,400 per year, but pay approximately \$13,100 of this in rent annually, leaving a significantly lower discretionary budget. Renters also report the lowest wellbeing among all tenure types, even after controlling for income. This suggests that the financial and psychological burden of ongoing housing costs in retirement is substantial and is not fully compensated by lower total spending. This well-being gap will have interesting consequences for asset allocation decisions, especially as it pertains to whole-life allocation and spending strategies. One of the peculiarities of the New Zealand KiwiSaver system is that it allows for the use of accumulated savings to be withdrawn for the purchase of a first home. Balancing the competing goals of homeownership and retirement savings is a unique challenge for New Zealand investors, and one that may have important implications for their lifetime asset allocation decisions. For instance, if utility gained from homeownership is sufficiently high, it may justify depleting the investors retirement savings while young to achieve homeownership, even if this results in a lower savings balance at retirement. Inversely, if buying a house depletes the investor’s retirement savings to a degree that the excess burden of housing costs would have otherwise have been more than offset by the additional savings balance at retirement, it may be optimal to delay homeownership and instead prioritise retirement savings.

The importance of homeownership is expected to increase over time. Retirement Income Interest Group (2024) note that home ownership among New Zealanders aged 65 and over is projected to fall from around 80 percent today to approximately 60 percent by 2048, as younger cohorts face higher barriers to entry in the housing market. This shift implies that expenditure benchmarks calibrated on the current cohort of retirees (who benefit from unusually high ownership rates) will understate spending requirements for future generations, particularly with respect to housing costs.

3.2.4 Expenditure Benchmarks for New Zealand Retirees

The most widely used empirical benchmarks for New Zealand retirement spending are the annual Retirement Expenditure Guidelines published by the NZ Fin-Ed Centre at Massey University (Matthews, 2025). These are derived from the Statistics New Zealand Household Economic Survey (HES) and reported at two tiers: a “No Frills” budget based on the second income quintile of retired HES households, reflecting a basic standard of living with few luxuries, and a “Choices” budget based on the fourth quintile, reflecting a more comfortable lifestyle with some discretionary spending. The 2025 figures, adjusted to 30 June 2025, are shown in Table 2.

Table 2: New Zealand Retirement Expenditure Guidelines, 2025. Weekly spending, with after-tax NZ Super rates for comparison. Source: Matthews (2025).

Budget	One-person household		Two-person household	
	Metro	Provincial	Metro	Provincial
No Frills (\$/week)	705	581	937	1,061
Choices (\$/week)	791	772	1,780	1,243
NZ Super after tax (\$/week)	519 (single)		799 (couple)	

For the No Frills provincial single-person household, NZ Super alone covers approximately 89 percent of spending, implying a required annual drawdown of only around \$3,200 from private savings. For the No Frills metro single, the gap is somewhat larger at around \$9,700 per year. Couples at the Choices tier, particularly in metro areas, face substantially higher gaps; the Choices metro couple budget of approximately \$92,600 per year implies a drawdown requirement of around \$51,000 annually above NZ Super. However, this upper estimate should be interpreted cautiously, since the fourth income quintile of HES retired households likely includes a material proportion of households still receiving employment or investment income, rather than pure retirees drawing down accumulated savings.

A central limitation of these guidelines is that they are constructed from cross-sectional data and therefore implicitly treat retirement spending as fixed in real terms throughout retirement. The HES samples households of all retirement ages at a single point in time, which means the guidelines effectively average the higher spending of the more numerous newly retired with the lower spending of older cohorts, without capturing the within-person spending trajectory over time (Retirement Income Interest Group, 2024). If real spending declines by approximately 2 percent per year after age 65 (as both Retirement Income Interest Group (2024) and Le and Richardson (2023) find) then the guidelines will overstate the savings balance required at age 65 for any given income target.

Retirement Income Interest Group (2024) quantify this directly. Using a replacement-rate methodology, they calculate that reaching a total income target of \$56,000 per year in real terms for a single retiree, on top of NZ Super, over a retirement from age 65 to 90, requires a savings balance of \$605,000 under a constant-real-spending assumption. If instead nominal income is assumed to remain flat (equivalent to a 2 percent real decline when inflation is also 2 percent), the required balance falls to \$375,000, a reduction of 40 percent. An intermediate assumption of a 1 percent real annual decline yields a balance of \$480,000. These calculations assume a 3.5 percent net investment return, which is not particularly high and is broadly consistent with the long-run real return on a relatively conservative portfolio.

For the purposes of this study, we treat the expenditure guidelines as an empirical anchor for age-65 spending levels. We calibrate initial retirement spending to the No Frills and Choices benchmarks as the lower and upper bounds for a representative retiree, and model real consumption as declining at approximately 2 percent per year thereafter. NZ Super is treated as a guaranteed, inflation-linked income floor from age 65. Under this calibration, the required drawdown from financial assets diminishes progressively as retirement spending falls toward the level already covered by NZ Super. This pattern has material implications for both the required savings balance at retirement and the optimal asset allocation strategy during decumulation.

4 Methods for Simulating and Optimising Lifetime Wealth Trajectories

The crux of our analysis is the ability to simulate realistic wealth trajectories for individual investors. This in and of itself is not a novel contribution, but the specific features of our simulation framework are designed to accommodate the particular questions we seek to answer. The goal, however, is to develop a flexible, modular framework that can be extended beyond the specific questions of this thesis to support ongoing research and policy analysis for Consilium and others. In this respect, while the specific questions we ask in this thesis are narrowly focused on the optimality of age-based glidepaths in the New Zealand retirement context, the framework we develop is intended to be more broadly applicable to a wide range of questions in the lifetime asset allocation space, including questions about optimal whole-life allocation and spending strategies, the role of housing and other non-financial assets, and the interaction between retirement savings and other life goals such as homeownership. To this end, much of this section will describe the general structure of the simulation framework, with specific details about the calibration to the New Zealand context and the particular questions

we ask in this thesis reserved for later sections. Included also, in the appendix of this thesis will be possible extensions to the framework that could be implemented in future work, such as the inclusion of housing and other non-financial assets, and the modelling of homeownership decisions and their interaction with retirement savings and spending.

The problem of simulating lifetime wealth trajectories has essentially two components: the methods used to simulate investor behaviour and the methods used to simulate return dynamics. The former includes the specification of the asset allocation strategy, the spending rule, and the risk measure used to evaluate outcomes. The latter includes the choice of return-generation methodology (IID Monte Carlo, historical rolling periods, or block bootstrap). In this section, we will describe the general structure of our simulation framework, and how it accommodates different choices in these areas. We will also discuss the advantages and disadvantages of different approaches, and how we have calibrated our framework to the New Zealand context.

4.1 Lifecycle Simulation Framework

With the exception of the continuous-time approach pioneered by Merton (1969), the majority of the literature on simulating lifetime wealth trajectories uses discrete-time approaches that model the investor's wealth at regular intervals, such as annually or monthly (e.g. Anarkulova et al., 2025; Estrada, 2014; Mäkinen & Toivanen, 2024). This approach is far easier to implement and makes less assumptions about the underlying return process than the continuous-time approach, which relies on strong assumptions about the return process (for example, that returns are normally distributed and independent over time) that are not supported by empirical evidence.

In this framework, we evolve the investor's wealth over time according to a simple recursive transition equation that accounts for returns, contributions, and withdrawals. For the transition from time t to $t + 1$, the investor's wealth is updated as follows in (equation 1):

$$w_{t+1} = g_{t+1}w_t + \pi_{t+1} \quad (1)$$

Where w_t is the wealth at time t , g_{t+1} is the growth factor on the portfolio between time t and $t + 1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t + 1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Assuming that the portfolio is rebalanced at the end of each period, the growth factor can be calculated as follows in equation 2:

$$g_{t+1} = 1 + \mathbf{a}_t^\top \mathbf{r}_{t+1} \quad (2)$$

Where $\mathbf{a}_t$ is the asset allocation vector at time t , with each element $a_{i,t}$ representing the proportion of wealth invested in asset class i , and $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t + 1$. Both $\mathbf{a}_t$ and $\mathbf{r}_{t+1}$ are vectors of length k , where k is the number of asset classes in the portfolio.

In a Monte Carlo simulation, we generate a large number of possible investor wealth trajectories by simulating returns across multiple paths. The returns are stored in a tensor $\mathcal{R}$ of dimension $N \times T \times k$, where N is the number of simulated paths, T is the number of time periods, and k is the number of asset classes. Similarly, the allocation policy can be represented as a tensor $\mathcal{A}$ of dimension $N \times T \times k$, where each element $A_{i,t,j}$ represents the allocation to asset class j at time t for simulation path i . In our nomenclature, asset allocation policies are functions or sets of rules that determine the allocation vector $\mathbf{a}_t$ at each time t . By representing the allocation policy as a tensor, we can accommodate a wide range of policies, including those that depend on path-specific state variables such as wealth or age. For example, a simple age-based glidepath can be represented by setting all paths to have identical allocations at each time, while a more complex policy that adjusts allocations based on the investor's current wealth can be represented by allowing the allocations to vary across paths at each time step.

At each time $t + 1$, we extract the $N \times k$ matrix $\mathbf{R}_{:,t+1}$ containing returns at time $t + 1$ across all paths and asset classes, and the $N \times k$ matrix $\mathbf{A}_{:,t}$ containing allocations at time t for all paths. The growth factors for all paths are then computed in vector form as shown in equation 3:

$$\mathbf{g}_{t+1} = \mathbf{1}_N + \sum_{j=1}^k \mathbf{R}_{:,t+1,j} \odot \mathbf{A}_{:,t,j} \quad (3)$$

Where $\mathbf{1}_N$ is an $N \times 1$ vector of ones, $\mathbf{R}_{:,t+1,j}$ and $\mathbf{A}_{:,t,j}$ are the $N \times 1$ column vectors for asset class j , $\odot$ denotes element-wise multiplication, and $\mathbf{g}_{t+1}$ is the resulting $N \times 1$ vector of growth factors across all paths.

Equivalently, this can be expressed as $\mathbf{g}_{t+1} = \mathbf{1}_N + \text{rowsum}(\mathbf{R}_{:,t+1} \odot \mathbf{A}_{:,t})$, where the element-wise product is summed across the asset class dimension. The wealth evolution across all paths then follows equation 4:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \quad (4)$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t , and $\odot$ denotes element-wise (Hadamard) multiplication. The wealth across all paths over all time periods can be represented as an $N \times T$ matrix, $\mathbf{W}$, where each row corresponds to a single simulated path of wealth over time.

Note that the system of equations above is agnostic to the return-generation process. The returns tensor $\mathcal{R}$ is a flexible input that can be generated using any methodology. This allows us to test the robustness of our findings across different assumptions about return dynamics, such as using historical returns, bootstrapped returns, or model-simulated returns. Additionally, $\mathbf{a}_t$ and $\boldsymbol{\pi}_{t+1}$ can be specified as functions of time, wealth, or other state variables, allowing for a wide range of asset allocation and spending strategies to be simulated within the same framework. This flexibility is a key strength of our approach, as it allows us to explore a broad set of strategies and assumptions without needing to fundamentally alter the underlying simulation mechanics. It is on the construction of these functions and the calibration of the return-generation process that we will focus in the next sections.

Implications for Consilium: Matrix Operations and Computational Efficiency

Consilium has used a similar, discrete-time, Monte Carlo simulation framework in its proprietary “Goal Planning Software” (GPS) tool for some time. However, this tool has severe limitations in terms of its flexibility and computational efficiency. The GPS tool is built on a non-vectorised implementation that relies heavily on for-loops to evolve wealth trajectories across simulated paths. This approach is computationally inefficient, particularly as the number of paths and time periods increases, which limits the tool’s ability to support more complex strategies or larger simulations.

The effort taken to express the model in matrix form is not only for mathematical clarity; it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (for example, `numpy` (Harris et al., 2020)) to vectorise the the calculations across the matrices longest dimension (the path dimension). This means that we can compute the wealth evolution across all paths virtually simultaneously using efficient matrix operations, rather than iterating through each path sequentially with for-loops. This vectorised approach can take advantage of modern CPU architectures and optimised libraries to perform computations much faster than a non-vectorised implementation, which can substantially accelerate simulation when path counts and horizon lengths are large. Compared to the GPS, which takes upwards to 30 seconds to simulate less than 1000 paths over a 30-year horizon, our matrix-based implementation can simulate 100,000 paths over a 50-year horizon in under a second on a standard laptop. This dramatic improvement in computational efficiency enables us here to explore a much wider range of strategies with much more vigour than would be possible with the GPS tool, and also opens the door to more complex strategies that may require more computational resources, such as those that involve path-dependent allocations or more sophisticated risk measures.

Refactoring the GPS tool to use matrix operations would be a significant improvement in its computational efficiency and flexibility. It would allow Consilium to support more complex strategies and larger simulations, which could significantly enhance the tool’s value for financial advisors and their clients.

4.1.1 Modelling Contributions and Withdrawals

The flow of wealth into and out of the portfolio is critical not only for the evolution of system, but also for the evaluation of outcomes. During retirement, when our investors is dissaving to fund consumption, how much the investor is withdrawing and is able to withdraw is a key determinant of welfare. Modelling contributions and withdrawals in a realistic way is therefore essential for simulating realistic wealth trajectories and evaluating the performance of different asset allocation strategies. There have been many approaches to modelling and planning withdrawals in the literature and financial advice space. Perhaps the most widely known is the “4% rule” popularised by Bengen (1994), which suggests that a retiree can safely withdraw 4% of their initial retirement savings balance in the first year of retirement, and then adjust that amount for inflation in subsequent years, without running out of money over a 30-year retirement period.

In their report, Retirement Income Interest Group (2024) propose several alternative withdrawal rules tailored to the New Zealand context, each reflecting different assumptions about spending trajectories through retirement. We briefly describe these rules here in terms of the consumption at time t , denoted c_t .

The 6% Rule:

This rule specifies a fixed nominal withdrawal amount equal to 6% of the initial retirement balance:

$$c_t = 0.06 \times w_0 \quad (5)$$

This rule front-loads spending and is well-aligned with the empirical finding that real spending declines by approximately 2% per year after age 65. When general inflation is also 2%, holding nominal spending constant produces the observed real decline. However, because withdrawals are fixed in nominal terms while returns are stochastic, there is a risk that the fund will be depleted before death.

The Inflated 4% Rule:

This is a variant of Bengen’s original formulation, where the initial 4% withdrawal is adjusted annually for inflation:

$$c_t = 0.04 \times w_0 \times s_t \quad (6)$$

where s_t is the cumulative inflation factor at time t since retirement. This rule maintains constant real spending, which is inconsistent with the empirical evidence of declining consumption but is more conservative than the 6% rule. It is more likely to preserve wealth through retirement and may be appropriate for retirees with lower risk tolerance or longer expected lifespans.

The Fixed Date Rule:

This rule divides the current wealth balance by the number of years remaining until a specified terminal age T , at which point the retiree is assumed to rely solely on NZ Super:

$$c_t = \frac{w_t}{T - t} \quad (7)$$

This rule produces withdrawals that vary from year to year as wealth fluctuates with market returns. It maximises spending over the chosen horizon but does not account for longevity risk beyond the terminal age. The approach is simple to implement but may produce significant year-to-year volatility in consumption.

The Life Expectancy Rule:

This rule divides current wealth by the actuarial life expectancy at the retiree’s current age t :

$$c_t = \frac{w_t}{e_t} \quad (8)$$

where e_t is the remaining life expectancy at age t . This approach is dynamically efficient in the sense that, in expectation, it exhausts the fund exactly at death. Because life expectancy declines with age, the denominator shrinks over time, which tends to stabilise the withdrawal rate even as wealth declines. This is perhaps the most economically rational of the simple heuristics, as it balances current consumption against future needs in a way that accounts for mortality risk.

Required Minimum Distribution (RMD) Consumption:

Each of these rules is designed to be simple enough for retirees to implement without sophisticated planning tools, and each implicitly or explicitly responds to the empirical observation that real spending declines through retirement. Retirement Income Interest Group (2024) note that the 6% rule is consistent with the observed 2% annual real decline when general inflation is also 2%, as it holds nominal spending approximately constant (after the asset returns are accounted for).

However, none of these simple heuristics on its own is suitable for modelling the withdrawal behaviour (i.e. π_t) of a representative retiree in our simulation framework. Most obviously, retirees have minimum spending

needs that must be met regardless of market returns, and these needs are not captured by any of the rules above. For example, if a retiree has a minimum real spending requirement of \$30,000 per year to cover essential expenses, then the 6% rule would fail to meet this requirement if the initial retirement balance is less than \$500,000. Similarly, the Fixed Date Rule and Life Expectancy Rule could produce withdrawals that are too low to meet essential needs in years when wealth is depleted by poor market returns.

To model this more realistically, we specify a minimum real spending requirement or floor, $\underline{c}_t$, that must be met in each period. The withdrawal in each period is then the maximum of the amount specified by the chosen rule and the minimum spending requirement:

$$c_t = \max(\text{Rule}(w_t, t), \underline{c}_t) \quad (9)$$

Because this is set in the New Zealand context, our investors will also receive NZ Super, which provides a guaranteed, inflation-linked income floor from age 65. The income received from NZ Super is calculated as:

$$v_t = v_0 \times s_t \quad (10)$$

Where v_0 is the initial NZ Super payment at age 65, and s_t is the cumulative inflation factor at time t since retirement.

Additionally, we do not allow for our investors go into debt to meet their spending needs. If the desired withdrawal exceeds the current wealth balance, the investor simply withdraws all remaining wealth and then relies on NZ Super for the remainder of their income needs. This that the total consumption at time is the minimum of the amount desired by the investor and the amount that can be funded by the current wealth balance plus NZ Super:

$$\hat{c}_t = \min(c_t, w_t + v_t) \quad (11)$$

Net contributions to the portfolio are then given by:

$$\pi_t = -\hat{c}_t + v_t \quad (12)$$

Implications for Consilium: Modelling Withdrawals in GPS

The GPS tool currently models withdrawals using a simple fixed dollar amount that can be adjusted for inflation. This is a particularly inflexible approach that does not allow for the kind of dynamic withdrawal strategies described above. Since the GPS is a whole-life planning tool which models both the accumulation and decumulation phases, there is substantial heterogeneity in distribution of wealth at retirement across simulated paths in many scenarios. This means that a fixed withdrawal amount may be too high for some paths, leading to early depletion of the fund, and too low for other paths, leading to underconsumption relative to the retiree's needs. Modelling withdrawals in a more dynamic way, such as implementing the rules above, would allow the GPS to better capture the trade-offs that retirees face in balancing current consumption against future needs, and would also allow for a more realistic representation of the risks associated with different asset allocation strategies.

These are relatively simple rules to implement, and could be added as options in the GPS tool without requiring a fundamental overhaul of the underlying simulation framework. Implementing these rules would enhance the tool's ability to support more realistic retirement planning scenarios and provide more nuanced advice to clients about the risks and trade-offs associated with different withdrawal strategies.

4.1.2 Modelling Asset Allocation Policies

The asset allocation policy is the key decision variable in our analysis, and the one that we seek to optimise. Our policy function specifies the allocation to each asset class at each time t for each simulation path, and is denoted by the operator $\mathcal{A}$. New Zealand investors cannot leverage or short-sell, so we restrict our attention to policies that only allow for long positions in the available asset classes. Because short-selling and leverage are not allowed, we can reduce the dimensionality of the problem by treating one of the asset classes as the "safe" asset, with the allocation to this asset class determined residually as one minus the sum of the allocations to the other asset classes. For example, if we have two asset classes (a risky asset and a safe asset), we can represent the allocation to the risky asset as a_t and the allocation to the safe asset as $1 - a_t$. This reduces the number of decision variables in our optimisation problem and simplifies the search for optimal policies.

This means that the allocation vector $\mathbf{a}_t$ must satisfy the following constraints:

$$\sum_{j=1}^k a_{j,t} \leq 1 \quad \text{and} \quad a_{j,t} \geq 0 \quad \text{for all } j \quad (13)$$

As we’ve already noted, most of the literature on optimal asset allocation in retirement focuses on age-based glidepaths, which is a policy operator that specifies the allocation to each asset class as a function of the investor’s age (or equivalently, time since retirement (t)). In this case, the asset allocation at time t is given by:

$$\mathbf{a}_t = f(t; \boldsymbol{\theta}) \quad (14)$$

Where f is a function parameterised by a set of parameters $\boldsymbol{\theta}$. For example, a simple linear glidepath that starts with an equity allocation of 80% at retirement and declines to 20% at age 90 can be represented by a linear function with two parameters: the initial allocation and the final allocation. More complex functional forms, such as piecewise linear or spline functions, can also be used to capture more nuanced glidepath shapes. The key point is that by parameterising the asset allocation policy in this way, we can use optimisation techniques to search for the set of parameters $\boldsymbol{\theta}$ that maximises our chosen risk measure, which we will discuss in the next section.

We also wish to explore a more flexible class of policies that allow for path-dependence, where the allocation can depend on the investor’s current wealth w_t in addition to time. This allows for the possibility that the optimal asset allocation strategy may involve adjusting the allocation based on the investor’s current financial situation, which is a more realistic assumption than a purely age-based glidepath. Our formulation of a wealth-time dependent policy follows the control-matrix approach of Mäkinen and Toivanen (2024), where the allocation at time t is determined by a matrix of parameters that specify the allocation for different, discrete combinations of time and wealth levels. Continuous allocations are then obtained by interpolating between the discrete points with bilinear interpolation. This approach allows for a high degree of flexibility in the shape of the asset allocation policy, while still being tractable for optimisation. We like this approach because the control-matrix structure is relatively easy to understand and interpret, and the use of interpolation allows for smooth policies without requiring a large number of parameters. This is particularly important when we consider that the optimisation problem can become quite complex as the number of parameters increases, so having a parsimonious parameterisation is beneficial for both computational efficiency and interpretability.

To illustrate this more concretely, consider Figure 3, which shows a simple control matrix for a two-asset portfolio. The matrix has six time steps (0 to 5) and five wealth levels (0 to 4), giving a 6 by 5 grid. Each cell corresponds to a risky-asset allocation (with the remainder allocated to the safe asset). For instance, if an investor is at time step 2 with wealth level 3, the policy returns the allocation in that cell. For time or wealth values between grid points, we use the interpolation method proposed by Mäkinen and Toivanen (2024). For example, at time 2.5 and wealth 3.5, bilinear interpolation uses the four neighbouring grid points to determine allocation.

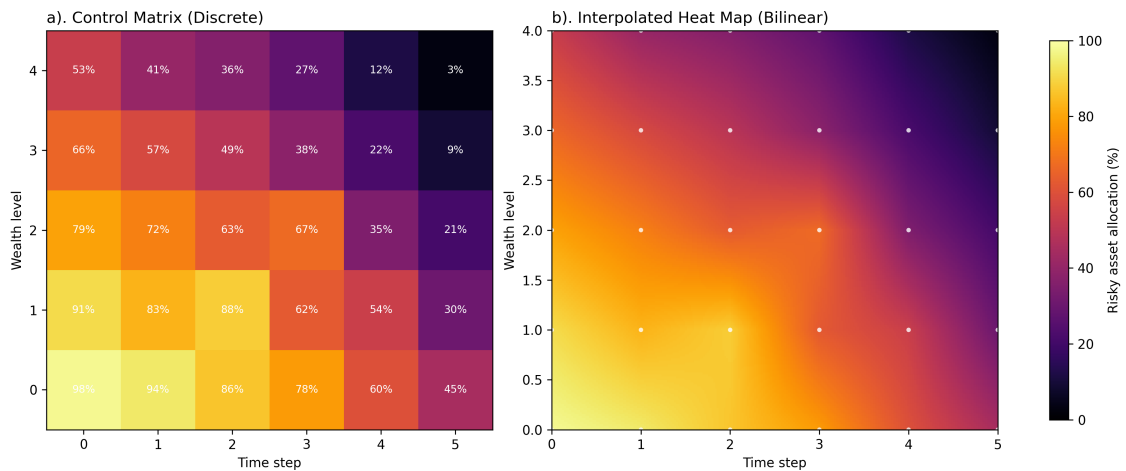


Figure 3: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

For allocations with more than two asset classes, the control matrix can be extended to a higher-dimensional tensor, with each additional dimension representing an additional asset class. Mäkinen and Toivanen (2024)

use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined grid points, ensuring a smooth transition between different allocation strategies.

Duarte et al. (2021) use a neural network to create a highly flexible policy function that integrates a wide range of state variables, including time, wealth, and various macroeconomic indicators. This approach allows for a very rich class of policies, but it also introduces challenges in terms of interpretability and the risk of overfitting, particularly given the synthetic nature of the asset return data used in the simulations. We think that this is a promising direction for future research, but for the purposes of this thesis we will focus on just the age-based and wealth-time dependent policies described above, as they represent the two most readily available information that an investor would have when making asset allocation decisions.

4.2 Methods for Simulating and Bootstrapping Asset Returns

Central to our analysis is the ability to simulate a large number of realistic asset return trajectories. As previously noted, there are two broad approaches to simulating returns: synthetic return generation using a parametric model, and bootstrapping from historical returns. The advantage of synthetic return generation is that it allows us to explore a wider range of possible return dynamics and also to create a larger number of return trajectories than would be possible with historical data alone. The disadvantage is that it relies on strong assumptions about the return-generating process, which may or may not be an accurate representation of reality. For example, the easiest and therefore most commonly used model is the geometric Brownian motion (GBM) model, which assumes that returns are normally distributed and independent over time. In reality, asset returns exhibit features such as volatility clustering, fat tails, and serial correlation, which are not captured by the time-independent GBM model. Time series models such as GARCH can capture some of these features, but they also introduce additional complexity and make their own set of assumptions that may not hold in practice. Bootstrapping from historical returns, on the other hand, allows us to capture the empirical distribution of returns without making strong parametric assumptions. However, it is limited by the length and quality of the historical data. Underpinning both approaches is the assumption that the future will be similar to the past: in the case of synthetic return generation, this is because the model parameters are typically estimated or evaluated against historical data, and in the case of bootstrapping, this is because we are directly resampling from historical returns. This assumption may not hold if there are structural changes in the economy or financial markets that lead to different return dynamics in the future. This limitation is inherent to this type of research, and it forms the underlying uncertainty in any conclusions we draw from our simulations.

Consilium has historically used a simple geometric Brownian motion (GBM) model to generate synthetic returns for its analysis. Because the company already has established expectations of long-run returns and volatilities for different asset classes in its internal models, the GBM model is a convenient way to generate return trajectories that are consistent with these expectations. However, as noted above, the GBM model has significant limitations. For this reason, we have also implemented a bootstrapping approach that resembles the block bootstrap method proposed by Anarkulova et al. (2025), which allows us to capture the empirical distribution of returns, including features such as volatility clustering and fat tails, without making strong parametric assumptions.

4.2.1 Block Bootstrap Methodology

Our implementation follows the stationary block bootstrap framework of Politis and Romano (1994), adapted to the multi-asset context as in Anarkulova et al. (2025). The key insight of the block bootstrap is that by resampling contiguous blocks of historical returns rather than individual observations, we preserve the short-run time-series dependencies present in asset returns while still generating a large number of distinct synthetic paths.

The procedure works as follows. We begin with historical return data for three asset classes: short-term government bills (cash), intermediate-term government bonds, and international equities. For each simulated path of length T , we construct a return sequence by concatenating randomly selected blocks from the historical record. The block length ℓ is itself a random variable drawn from a geometric distribution with mean $\bar{\ell}$, which ensures that the resulting bootstrap series is stationary. Specifically, at each step we draw a block starting position uniformly from the historical data, then draw a block length $\ell \sim \text{Geometric}(1/\bar{\ell})$, and append the next ℓ observations to the simulated path. This process continues until we have accumulated T periods of returns.

Formally, let $\mathbf{R}^{\text{hist}}$ denote the $T_{\text{hist}} \times k$ matrix of historical returns, where T_{hist} is the number of historical periods and $k = 3$ is the number of asset classes (bills, bonds, equities). To generate a single simulated path of

length T , we repeat the following until the path is complete:

1. Draw a starting index $t_0 \sim \text{Uniform}\{1, 2, \dots, T_{\text{hist}} - \bar{\ell}\}$
2. Draw a block length $\ell \sim \text{Geometric}(1/\bar{\ell})$
3. Append returns $\mathbf{R}_{t_0}^{\text{hist}}, \mathbf{R}_{t_0+1}^{\text{hist}}, \dots, \mathbf{R}_{t_0+\ell-1}^{\text{hist}}$ to the simulated path

In the case that the block extends beyond the end of the historical data, we simply cut off the block at the end of the data and start a new block from a randomly selected starting index, effectively creating two blocks that are concatenated together.

The choice of mean block length $\bar{\ell}$ involves a trade-off. Longer blocks preserve more of the time-series structure, including medium-run dynamics such as business cycle effects and mean reversion, but reduce the diversity of simulated paths by limiting the number of distinct resampling configurations. Shorter blocks increase diversity but may fail to capture persistent features of return dynamics. Anarkulova et al. (2025) use a mean block length of 120 months (10 years) for their analysis of US equity and bond returns, arguing that this preserves economically relevant persistence while generating sufficient path diversity for robust optimisation. Following their approach, we calibrate our mean block length to approximately 10 years of annual data, giving $\bar{\ell} = 10$.

An important feature of the block bootstrap is that it naturally preserves the cross-sectional correlation structure among asset classes. Because we resample entire rows of the historical return matrix (i.e., contemporaneous returns across all assets), the bootstrap paths retain the empirical co-movement of bills, bonds, and equities. This is critical for retirement portfolio analysis, as the diversification benefits of holding multiple asset classes depend directly on their correlation structure during different market regimes.

For our analysis, we use three asset classes that represent the feasible investment universe for a typical New Zealand retail investor:

- **Domestic Cash (Bills):** Short-term government securities with minimal default risk and low volatility.
- **Domestic Bonds:** Intermediate-term government bonds that provide a higher expected return than cash but with greater volatility and some interest rate risk.
- **International Equities:** A diversified portfolio of global equity markets, providing exposure to long-run economic growth with substantial short-run volatility. Following Anarkulova et al. (2025), we construct this series as a market-capitalisation-weighted aggregate of developed-market equity indices.

We also aim to preserve the cross-sectional correlation of these assets against other key macroeconomic variables, in particular inflation. To do this, we include the domestic inflation rate as an additional variable in the historical data matrix, and block bootstrap it alongside the asset returns.

4.2.2 Data for Block Bootstrapping

The core dataset for macro-financial variables is the Jordà-Schularick-Taylor (JST) Macrohistory Database (Jordà et al., 2017), a comprehensive panel of annual macroeconomic and financial data covering 18 advanced economies from 1870 to the present. The database was developed to provide researchers with long-run time series that span multiple financial cycles, wars, and regime changes, making it particularly valuable for studying asset return dynamics over horizons relevant to retirement planning.

The JST database includes total returns (capital gains plus reinvested dividends) for equities, bonds, and bills, alongside inflation (CPI), exchange rates, GDP, credit aggregates, and house prices. Importantly, the asset return series are constructed consistently across countries using a common methodology, which facilitates international comparison and aggregation. For equity returns, the database draws on a combination of national stock market indices; for bonds, it uses long-term government securities; and for bills, it uses short-term government debt or comparable money-market instruments. All series are reported in nominal local currency terms, with separate CPI series provided to compute real returns.

For our analysis, we extract annual data for the 18 countries in the JST database (Australia, Belgium, Canada, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, and United States) covering the period 1950–2020. We focus on the post-1950 period to avoid the structural breaks associated with World War II and to ensure that the return dynamics reflect the modern financial system with which current retirees are familiar. This gives us $T_{\text{hist}} = 71$ annual observations per country, or 1,278 country-year observations in total when pooled across all 18 economies.

To construct our three asset class return series, we proceed as follows:

5 Optimal Lifetime Asset Allocation Strategies for New Zealand Retirees

Having established the simulation and optimisation framework, we now apply it to evaluate candidate lifetime allocation policies in the New Zealand retirement context.

5.1 Optimal Age-Based Glide-Paths

5.2 Optimal Wealth-Responsive Allocation Strategies

6 Discussion and Implications

7 Conclusion

References

- Aguiar, M., & Hurst, E. (2008). *Deconstructing lifecycle expenditure* (tech. rep. No. 13893). National Bureau of Economic Research. <https://doi.org/10.3386/w13893>
- Alonso-García, J., Bateman, H., Bonekamp, J., van Soest, A., & Stevens, R. (2022). Saving preferences after retirement. *Journal of Economic Behavior & Organization*, 198, 409–433. <https://doi.org/10.1016/j.jebo.2022.04.005>
- Anarkulova, A., Cederburg, S., & O’Doherty, M. S. (2025, March). Beyond the status quo: A critical assessment of lifecycle investment advice [SSRN Working Paper]. <https://doi.org/10.2139/ssrn.4590406>
- Ando, A., & Modigliani, F. (1963). The “life cycle” hypothesis of saving: Aggregate implications and tests. *The American Economic Review*, 53(1), 55–84. <https://www.jstor.org/stable/1817129>
- Australian Taxation Office. (n.d.). *Super guarantee percentage* [Retrieved 6 July 2023].
- Banks, J., Blundell, R., Levell, P., & Smith, J. P. (2016). *Life-cycle consumption patterns at older ages in the us and the uk: Can medical expenditures explain the difference?* (Tech. rep. No. 22513). National Bureau of Economic Research. <https://doi.org/10.3386/w22513>
- Barrett, G., & Brzozowski, M. (2010). *Involuntary retirement and the resolution of the retirement-consumption puzzle: Evidence in australia* (tech. rep.). University of Sydney, School of Economics.
- Battistin, E., Brugiavini, A., Rettore, E., & Weber, G. (2009). The retirement consumption puzzle: Evidence from a regression discontinuity approach. *American Economic Review*, 99(5), 2209–2226. <https://doi.org/10.1257/aer.99.5.2209>
- Becker, G. S. (1965). A theory of the allocation of time. *The Economic Journal*, 75(299), 493–517. <https://doi.org/10.2307/2228949>
- Bengen, W. P. (1994). Determining withdrawal rates using historical data. *Journal of Financial Planning*, 7(4), 171–180. <https://www.financialplanningassociation.org/sites/default/files/2021-04/MAR04%20Determining%20Withdrawal%20Rates%20Using%20Historical%20Data.pdf>
- Bengen, W. P. (1996). Asset allocation for a lifetime. *Journal of Financial Planning*, 9(4), 86–92. <https://www.financialplanningassociation.org/article/journal/AUG96-asset-allocation-lifetime>
- Blanchett, D. M. (2007). Dynamic allocation strategies for distribution portfolios: Determining the optimal distribution glide path. *Journal of Financial Planning*, 20(12), 68–81.
- Bodie, Z., Kane, A., & Marcus, A. J. (2023). *Investments*. McGraw-Hill US Higher Ed ISE. <https://ebookcentral.proquest.com/lib/canterbury/detail.action?docID=7190049>
- Bodie, Z., Merton, R. C., & Samuelson, W. F. (1992). Labor supply flexibility and portfolio choice in a life cycle model. *Journal of Economic Dynamics and Control*, 16(3), 427–449. [https://doi.org/https://doi.org/10.1016/0165-1889\(92\)90044-F](https://doi.org/https://doi.org/10.1016/0165-1889(92)90044-F)
- Campbell, J. Y. (1996). Understanding risk and return. *Journal of Political Economy*, 104(2), 298–345.
- Choi, J. J. (2022). Popular personal financial advice versus the professors. *Journal of Economic Perspectives*, 36(4), 167–192. <https://www.jstor.org/stable/27171135>
- De Nardi, M., French, E., & Jones, J. B. (2010). *Differential mortality, uncertain medical expenses, and the saving of elderly singles* (tech. rep. No. 16453). National Bureau of Economic Research. <https://doi.org/10.3386/w16453>
- Dexter, G. (2025, November 24). *Kiwisaver provider calls for increased contribution to be compulsory*. Retrieved December 6, 2025, from <https://www.rnz.co.nz/news/political/579777/kiwisaver-provider-calls-for-increased-contribution-to-be-compulsory>

- Dolvin, S. D., Templeton, W. K., & Rieber, W. (2010). Asset allocation for retirement: Simple heuristics and target-date funds. *Scholarship and Professional Work - Business*, (17). https://digitalcommons.butler.edu/cob_papers/17
- Duarte, V., Fonseca, J., Goodman, A. S., & Parker, J. A. (2021, December). *Simple allocation rules and optimal portfolio choice over the lifecycle* (Working Paper No. 29559). National Bureau of Economic Research. <https://doi.org/10.3386/w29559>
- Estrada, J. (2014). The glidepath illusion: An international perspective. *The Journal of Portfolio Management*, 40(4), 52–64. <https://doi.org/10.3905/jpm.2014.40.4.052>
- Estrada, J. (2015). The retirement glidepath: An international perspective. *The Journal of Investing*, 25(2), 28–36. <https://doi.org/10.3905/joi.2016.25.2.028>
- Felix, B. (2025). The most controversial paper in finance [Accessed: 2026-05-05]. https://www.youtube.com/watch?v=-nPon8Ad_Ug
- Financial Markets Authority. (2015, November 1). *Regulatory impact statement: Guidance on risk indicators and descriptions of managed funds* [PDF document]. Retrieved May 4, 2026, from <https://www.fma.govt.nz/assets/RIS/Regulatory-Impact-Statement-guidance-on-risk-Indicators-and-descriptions-of-managed-funds.pdf>
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2005). There is a risk-return trade-off after all. *Journal of Financial Economics*, 76(3), 509.
- Gilbert, A. (2025, November 27). *Lifting kiwisaver contributions to 12% makes sense – when the whole scheme is fixed*. Retrieved December 6, 2025, from <https://www.nzherald.co.nz/nz/lifting-kiwisaver-contributions-to-12-makes-sense-when-the-whole-scheme-is-fixed>
- Grattan Institute. (2019). *The savings behaviour of older households: Submission 435 – supplementary submission to inquiry into the implications of removing refundable franking credits* (Supplementary submission No. 435) (Draws from: Daley, J., Coates, B., Wiltshire, T., Emslie, O., Nolan, J., and Chen, T. (2018), “Money in Retirement: More than Enough”, Grattan Institute). Parliament of Australia.
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Hurd, M. D. (1987). Savings of the elderly and desired bequests. *The American Economic Review*, 77(3), 298–312. <https://www.jstor.org/stable/1804093>
- Hurd, M. D., & Rohwedder, S. (2003). *The retirement-consumption puzzle: Anticipated and actual declines in spending at retirement* (tech. rep. No. 9586). National Bureau of Economic Research. <https://doi.org/10.3386/w9586>
- Ibbotson, R., & Kaplan, P. (2001). Does asset allocation policy explain 40, 90, 100 percent of performance? *Yale School of Management, Yale School of Management Working Papers*, 56. <https://doi.org/10.2469/faj.v56.n1.2327>
- Jordà, Ò., Schularick, M., & Taylor, A. M. (2017). Macrofinancial history and the new business cycle facts. *NBER Macroeconomics Annual*, 31(1), 213–263. <https://doi.org/10.1086/690241>
- King, M. A. (1982). Asset holdings and the life-cycle. *The Economic Journal*, 92(366), 247–267. <https://doi.org/10.2307/2232439>
- Le, T., & Richardson, E. (2023, August). *Expenditure patterns of new zealand retiree households* (Motu Working Paper No. 23-07). Motu Economic and Public Policy Research. <https://motu.nz/our-research/population-and-labour/individual-and-group-outcomes/expenditure-patterns-nz-retiree-households>
- Love, D. A., Palumbo, M. G., & Smith, P. A. (2009). The trajectory of wealth in retirement. *Journal of Public Economics*, 93(1), 191–208. <https://doi.org/https://doi.org/10.1016/j.jpubeco.2008.09.003>
- Lundblad, C. (2007). The risk return tradeoff in the long run: 1836–2003. *Journal of Financial Economics*, 85(1), 123–150. <https://doi.org/10.1016/j.jfineco.2006.06.003>
- Luxon, C. (2025, November 23). *National to lift kiwisaver contributions*. Retrieved December 6, 2025, from <https://www.national.org.nz/news/20251123-hon-christopher-luxon-national-to-lift-kiwisaver-contributions>
- Mäkinen, R. A. E., & Toivanen, J. (2024). Monte carlo expected wealth and risk measure trade-off portfolio optimization. *SIAM Journal on Financial Mathematics*.
- Matthews, C. (2025, October). *New zealand retirement expenditure guidelines 2025* (Annual Report). NZ Fin-Ed Centre, Massey University. <https://www.massey.ac.nz/documents/1554/new-zealand-retirement-expenditure-guidelines-2025.pdf>

- Merton, R. C. (1969). Lifetime portfolio selection under uncertainty: The continuous-time case. *The Review of Economics and Statistics*, 51(3), 247–257. <https://doi.org/10.2307/1926560>
- Ministry of Social Development. (2022). *Description of new zealand’s current retirement income policies* (tech. rep.) (Background paper prepared for the Retirement Commissioner’s 2022 Review of Retirement Income Policy). Ministry of Social Development. Retrieved March 17, 2026, from <https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/2022-RRIP/Description-of-New-Zealands-Current-Retirement-Income-Policies-MSD-for-2022-RRIP-.pdf>
- Modigliani, F. (1986). Life cycle, individual thrift, and the wealth of nations. *The American Economic Review*, 76(3), 297–313. Retrieved May 5, 2026, from <http://www.jstor.org/stable/1813352>
- Modigliani, F., & Brumberg, R. (2005, September). Utility analysis and the consumption function: An interpretation of cross-section data. In *The collected papers of franco modigliani, volume 6*. The MIT Press. <https://doi.org/10.7551/mitpress/1923.003.0004>
- New Zealand Government. (n.d.). *Paying for residential care*. Retrieved April 13, 2026, from <https://www.govt.nz/browse/health/rest-homes-and-residential-care/pay-for-residential-care/>
- New Zealand Parliament. (2001). New Zealand superannuation and retirement income Act 2001 [Version as at 1 April 2026]. Retrieved May 4, 2026, from <https://www.legislation.govt.nz/act/public/2001/0084/latest/whole.html>
- OECD. (2025). *Pension markets in focus 2025*. OECD Publishing. Paris. <https://doi.org/10.1787/b095d0a0-en>
- Palmer, R. (2025, September 7). *Watch: Peters promises compulsory 10% kiwisaver and tax cuts*. Retrieved December 6, 2025, from <https://www.rnz.co.nz/news/national/572352/watch-peters-promises-compulsory-10-percent-kiwisaver-and-tax-cuts>
- Palumbo, M. G. (1999). Uncertain medical expenses and precautionary saving near the end of the life cycle. *The Review of Economic Studies*, 66(2), 395–421. <https://doi.org/10.1111/1467-937X.00092>
- Pfau, W. D., & Kitces, M. E. (2014). Reducing retirement risk with a rising equity glide path. *Journal of Financial Planning*, 27(1), 38–45. <https://www.financialplanningassociation.org/article/journal/JAN14-reducing-retirement-risk-rising-equity-glide-path>
- Politis, D. N., & Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313. Retrieved May 17, 2026, from <http://www.jstor.org/stable/2290993>
- Retirement Commission Te Ara Ahunga Ora. (2022, August 15). *New research highlights how reliant older new zealanders are on nz super*. Retrieved March 17, 2026, from <https://retirement.govt.nz/news/latest-news/new-research-highlights-how-reliant-older-new-zealanders-are-on-nz-super>
- Retirement Income Interest Group. (2024, December). *Spending patterns through retirement: Implications for retirement planning and drawdown* (Report). New Zealand Society of Actuaries. <https://actuaries.org.nz/content/uploads/2024/12/Spending-through-retirement-RIIG-Dec2024.pdf>
- Shiller, R. (2005). Life-cycle portfolios as government policy. *The Economists’ Voice*, 2(1), 14–14. <https://doi.org/10.2202/1553-3832.1095>
- Shiller, R. J. (2006). Life-cycle personal accounts proposal for social security: An evaluation of president bush’s proposal. *Journal of Policy Modeling*, 28(4), 427–444. <https://doi.org/https://doi.org/10.1016/j.jpolmod.2005.10.010>
- SuperLife. (2026, March 26). *Statement of investment policy and objectives (sipo): Superlife invest*. Retrieved March 31, 2026, from https://www.superlife.co.nz/files/LegalDocs/SIPO/SuperLifeInvest-Statement_of_Investment_Policy_and_Objectives.pdf
- Swoboda, K. (2014, March 11). *Major superannuation and retirement income changes in australia: A chronology* (Research Papers 2013–14). Australian Parliamentary Library.
- Te Ara Ahunga Ora: Retirement Commission. (2021). *Review of retirement income: Policy paper 03* (Policy Paper No. 03). Te Ara Ahunga Ora: Retirement Commission. <https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/TAAO-RC-Policy-Paper-2021-03.pdf>
- The Treasury. (2025, September 24). *He tirohanga mokopuna 2025: Statement on the long term fiscal position* (Retrieved from <https://www.treasury.govt.nz/publications/ltfp/he-tirohanga-mokopuna-2025> on December 6, 2025). The Treasury. New Zealand Government. Retrieved December 6, 2025, from <https://www.treasury.govt.nz/publications/ltfp/he-tirohanga-mokopuna-2025>
- Work and Income New Zealand. (2026a, April 1). *Benefit rates from April 2026*. Retrieved May 4, 2026, from <https://www.workandincome.govt.nz/products/benefit-rates/benefit-rates-april-2026.html>
- Work and Income New Zealand. (2026b, April 1). *Superannuation rates from April 2026*. Retrieved May 4, 2026, from <https://www.workandincome.govt.nz/eligibility/seniors/superannuation/how-much-you-can-get.html>

A Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)

Mäkinen and Toivanen (2024) formulate their Monte Carlo optimisation in a way that is similar to the framework described in the main text. As in our setup, they use a control matrix that allows asset allocation to respond jointly to age and wealth. Their objective function is based on terminal wealth moments, specifically variance or semi-variance. They then compute gradients of this objective with respect to control-matrix parameters using adjoint methods, which enables efficient gradient-based optimisation (in their case, a quasi-Newton method with BFGS updates).

Because this method is mathematically elegant and computationally efficient, we have attempted to replicate it here. The code for this replication can be found in the accompanying repository, and the results are shown in Figure 4 for the mean-variance and mean-semi-variance cases.

In line with the setup described by Mäkinen and Toivanen (2024), the replication uses a Monte Carlo horizon of $T = 20$ years with time step $\Delta t = 0.25$, giving $N = 80$ time steps. The policy is represented on a time-wealth control grid with $M = 31$ wealth nodes, upper wealth bound $W_{\max} = 30$, and bilinear interpolation between nodes. In the notebook, wealth is clipped to $[0, W_{\max}]$ before interpolation. Optimisation is performed with L-BFGS-B (a quasi-Newton method) with box constraints.

For the one-risky-asset, no-leverage case used in Figure 4, the full parameterisation is summarised in Table 3.

Table 3: Parameter setup for the one-risky-asset mean-variance replication.

Parameter	Value
Risk-free rate (r)	0.03
Risky-asset drift (μ_1)	0.0795
Risky-asset volatility (σ_1)	0.15
Contribution rate (π)	0.1
Initial wealth (w_0)	1
Minimum risky allocation ($p_{\min}$)	0
Maximum risky allocation ($p_{\max}$)	1
Number of simulation paths (K)	10^5
Risk-aversion weight (λ)	0.1

The replication results are broadly consistent with those reported by Mäkinen and Toivanen (2024). However, they also highlight important limitations. First, as an objective for a retirement investor, terminal-wealth variance is not especially intuitive. In our replication, the terminal-wealth distribution becomes bimodal because the optimiser shifts to a highly conservative policy at high wealth levels, truncating the upper tail that would normally appear under a log-normal-like process. Although this behaviour is mathematically consistent with variance minimisation, it is not obviously aligned with investor welfare. A similar pattern appears under semi-variance, albeit at higher wealth levels. Because semi-variance is computed relative to the mean, higher-return strategies can increase the penalty by raising the mean itself, again encouraging upper-tail truncation. This reinforces the need for risk measures that are both interpretable and behaviourally relevant.

Additionally, the control matrix has limited effective resolution in early periods: many simulated paths map to only a few low-wealth nodes because initial wealth is far below the fixed upper wealth bound. This reduces the optimiser’s ability to distinguish trajectories precisely when policy differences likely matter most due to compounding. It also creates many redundant parameters at wealth levels that are never reached (upper-left region in Figure 4). For this reason, we propose calibrating the control-matrix upper bound to the simulated wealth distribution so that resolution is concentrated where paths actually lie.

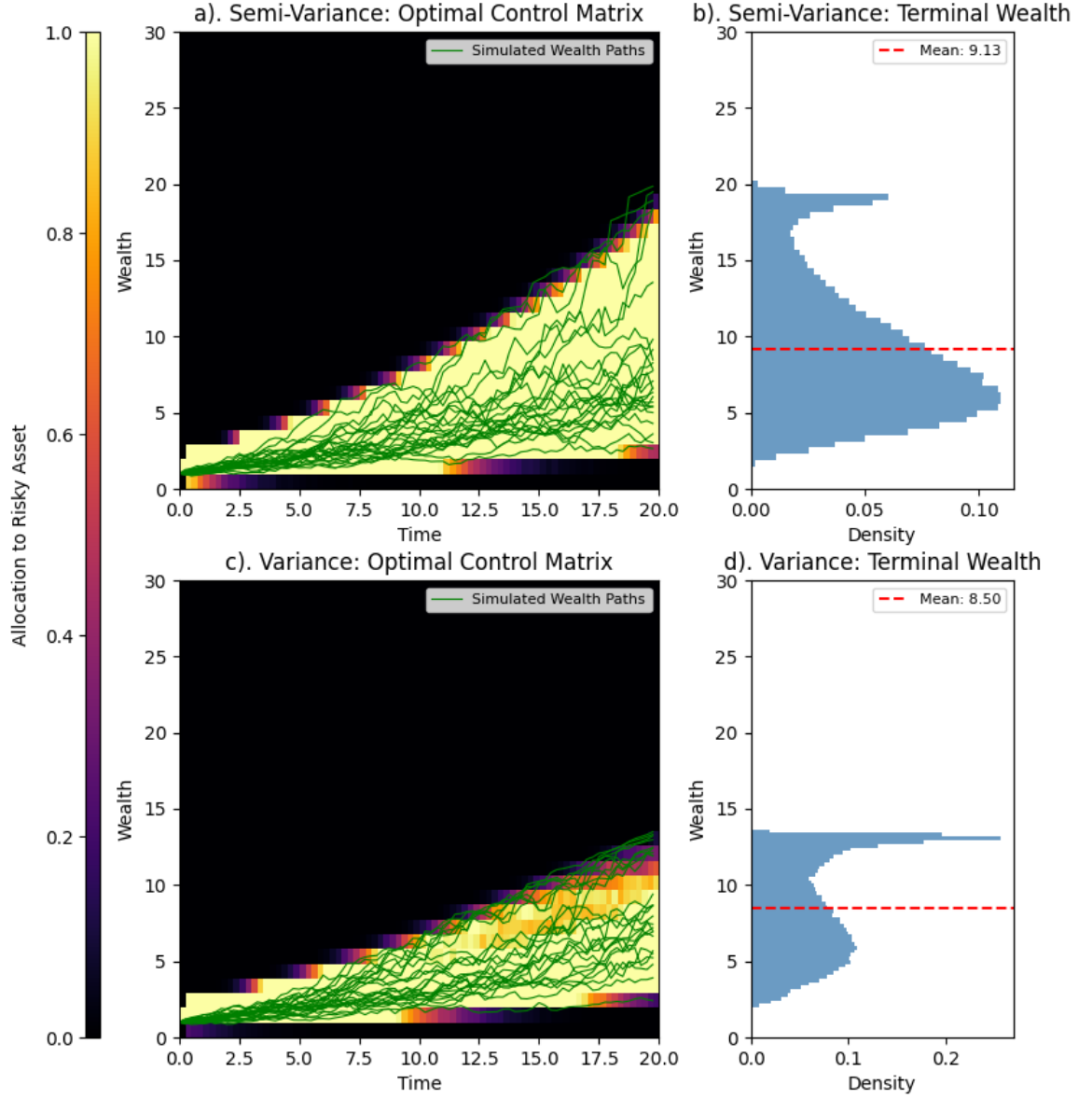


Figure 4: Replication of the control matrix optimization method proposed by Mäkinen and Toivanen (2024) for one risky asset and no leverage. The left column shows the optimal control matrix for the mean-semi-variance case (top) and the mean-variance case (bottom). The right column shows the corresponding histograms of terminal wealth for each risk measure respectively.